

TECHNICAL BLUEPRINT & SOFTWARE
SPECIFICATIONS

Elle.Be.O Growth Studio

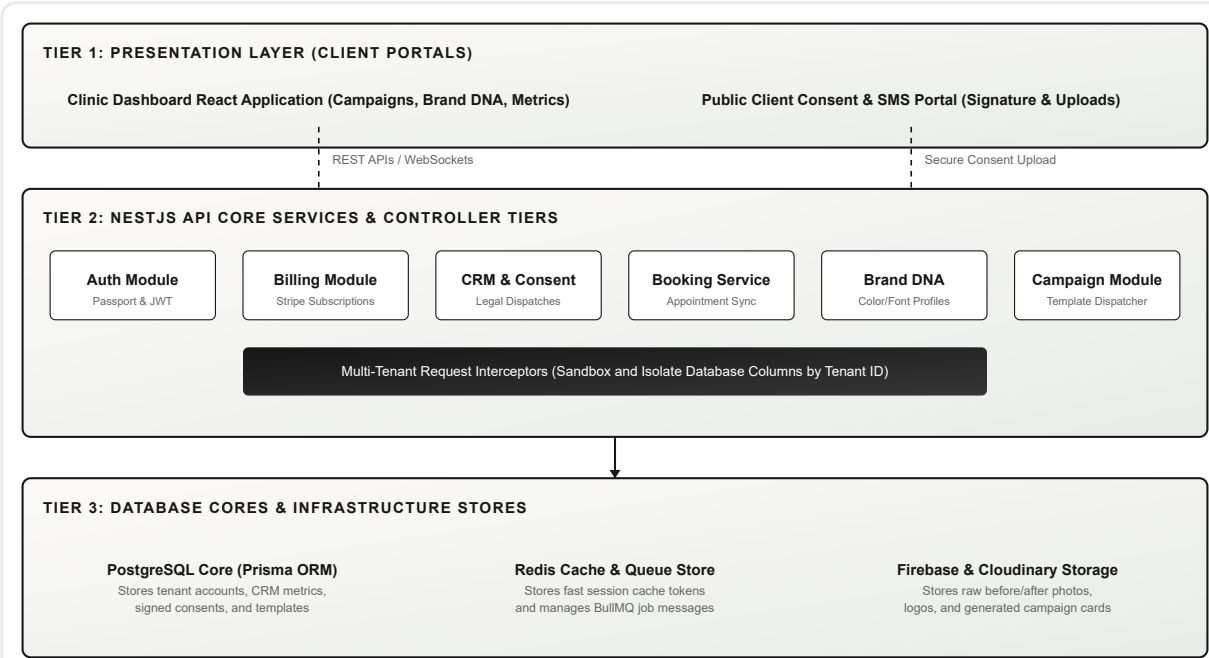
Full Application Architecture

Detailed multi-tier platform specification and specialized generative AI graphic compositing mapping, designed for technical teams and creative leads.

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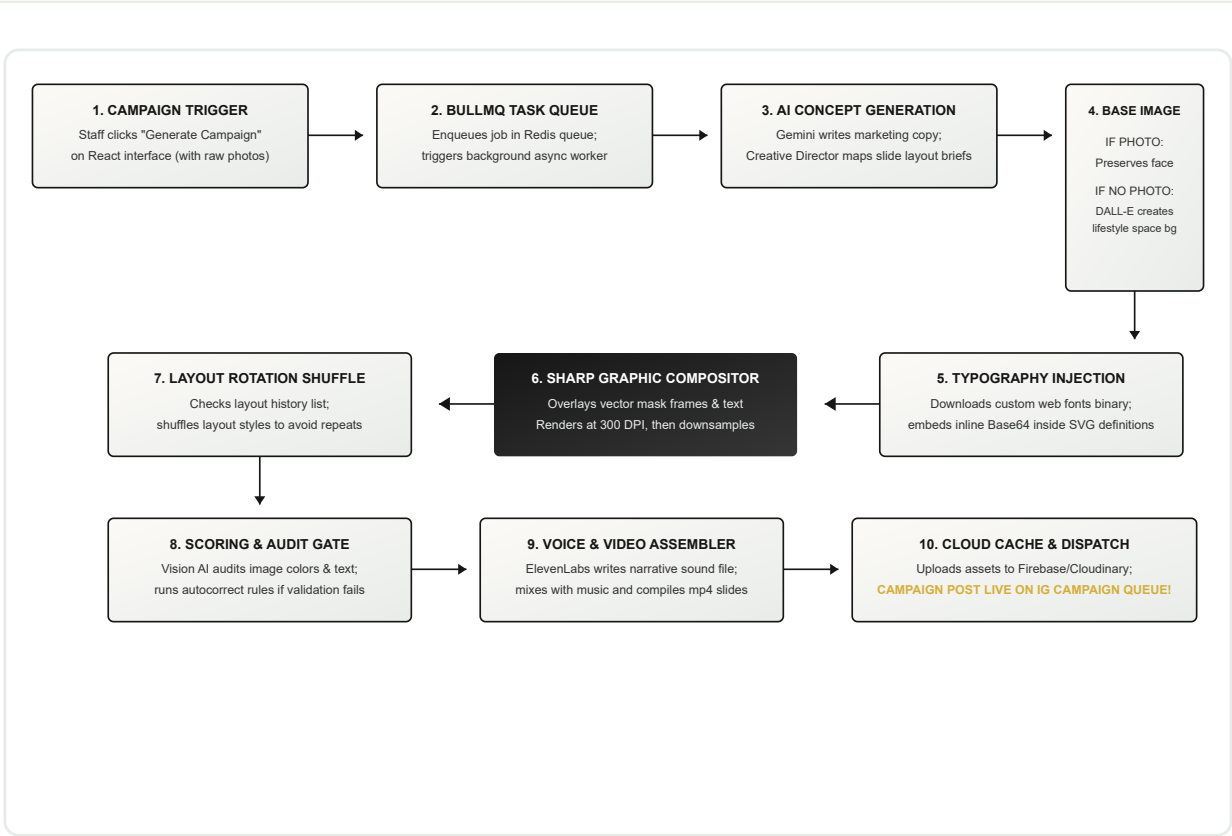
1. Platform Infrastructure & Tiers Blueprint

The core infrastructure manages secure multi-clinic accounts, Stripe subscription access gates, CRM client consent states, and appointment scheduling. The system partitions all database rows and media storage files to guarantee that each studio operates inside a fully sandbox environment.



2. End-to-End AI & Media Compositor Pipeline

This details the complete, end-to-end processing pipeline, from the moment a salon staff member initiates a campaign to the final high-definition media file export and social dispatch.



3. A Salon Client's Journey: Architecture in Action

To fully grasp how the backend codebase coordinates, let's follow the step-by-step technical execution during a normal business day at a medical aesthetics clinic.

Step 1: Booking & CRM Registration (Frontend & Postgres)

A client visits the clinic website and books a "Lip Filler Treatment."

- The **Presentation Layer** sends the booking request to the backend.
- The backend creates an entry in the **PostgreSQL datastore** (mapped via Prisma) mapping the Client ID, Tenant ID, and Booking Category.

Step 2: Legal Consent Guard (Auth & public-consent)

Before any staff member can snap a before/after treatment photo, legal consent is required.

- The **CRM Consent Service** triggers a Twilio SMS dispatch to the client's phone containing a secure link.
- The client signs their signature on the **Public Client Consent React Portal**.
- The database updates `consentApproved = true`. The **Consent Guard** on the server now unlocks photo uploads.

Step 3: Campaign Initiation & AI Ideation (BullMQ & chains)

At the end of the treatment, the staff uploads a raw clinic photo of the Lip Filler outcome. The server starts the generation job:

- Because generating graphics, texts, and videos takes heavy computational power, the server doesn't process it immediately. Instead, it drops the payload into the **BullMQ Background Task Queue** backed by **Redis cache**.
- An isolated background worker picks up the job. It passes the clinic Brand DNA guidelines to **Gemini / OpenAI chains** which write a high-end educational script tailored to "Lip Fillers" using the clinic's brand voice.

Step 4: Design Rasterization & Composition (Sharp, SVGs, Google Fonts)

The text captions are written, but we need high-end graphic cards.

- The **Brand DNA service** fetches the clinic's preferred custom typography (e.g. Playfair Display). It downloads Google Fonts binaries, converts them to base64, and dynamically embeds them into an SVG layout template.
- The **Sharp Compositor** reads the raw treatment photo, resizes it to high-definition standards, and overlays the vector SVG frame and text at **300 DPI high density**.
- If no photo was uploaded, the compositor fallback triggers, using DALL-E to generate gorgeous lifestyle backgrounds matching the primary and secondary branding colors (e.g., travertine stone details, wellness chairs) and overlays the brand text.

Step 5: Quality Gate Analysis (scoring-gate)

The graphics are built, but they must pass audit:

- The **Scoring Gate service** uses vision AI models to scan the generated images, checking for spelling errors, color clashes, or logo overlaps. If it fails, it issues automated correction directives. If it passes, the slide URLs are saved to Firebase Storage.

Step 6: Voiceover & Video Assembly (shotstack & elevenlabs)

For video posts (Reels/Stories):

- The **ElevenLabs integration** converts the AI strategy script into a professional human-sounding audio track.
- The **Reel Assembler service** synchronizes the audio voiceover with the visual slides and background music tracks, and compiles the final video file.